eedi-mining-misconceptions-in-mathematics

Private 7th (Public 2nd) Solution (Masaya Part)

Rank	7
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Topic ID	551430
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551430
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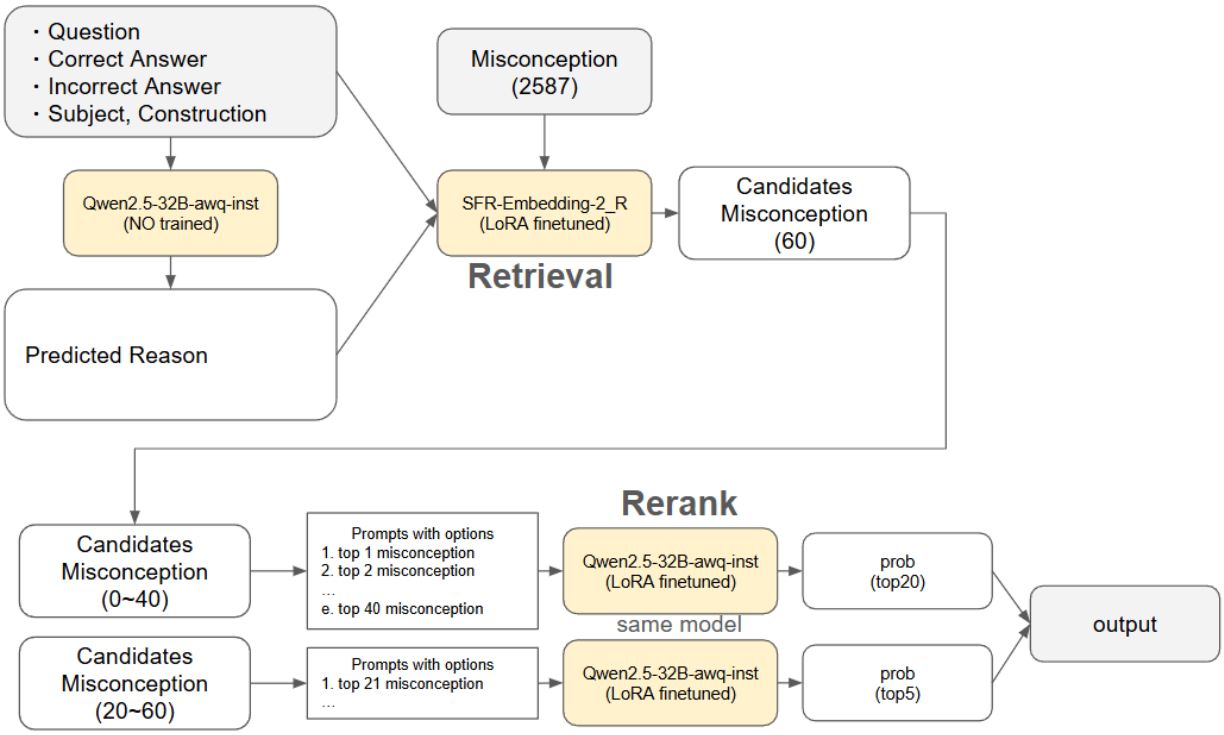
First of all, I would like to thank the organizers for hosting such an interesting competition and the teammate. (@kaerunantoka , @tereka and @aerdem4)

I enjoyed this competition because of interesting discussion with the teammate and exciting topic.

Overview of our team's solution is here.

I would like to share my part here.

Overview



Retrieval-Rerank pipeline

This competition is retrieval and rerank task.

So I made this pipeline as well as other participant may did.

Predicted Reason as Context by LLM (For Retrieval)

I generate the reasoning behind the incorrect answer.

I am inspired by this paper

Synthetic data for null (unknown) Misconception (For Retrieval)

I filled unknown Misconception with LLM.

I tried also rephrasing misconception but the recall is almost same.

Multi-Option Reranking

I make the Misconception candidates from the retriever to input the multi-option prompts of LLM in order to predict reranking once for each Question-Answer.

As a result, my pipeline finished in 210min (if predicted once, 140min).

Using vLLM is the idea by kaerururu.

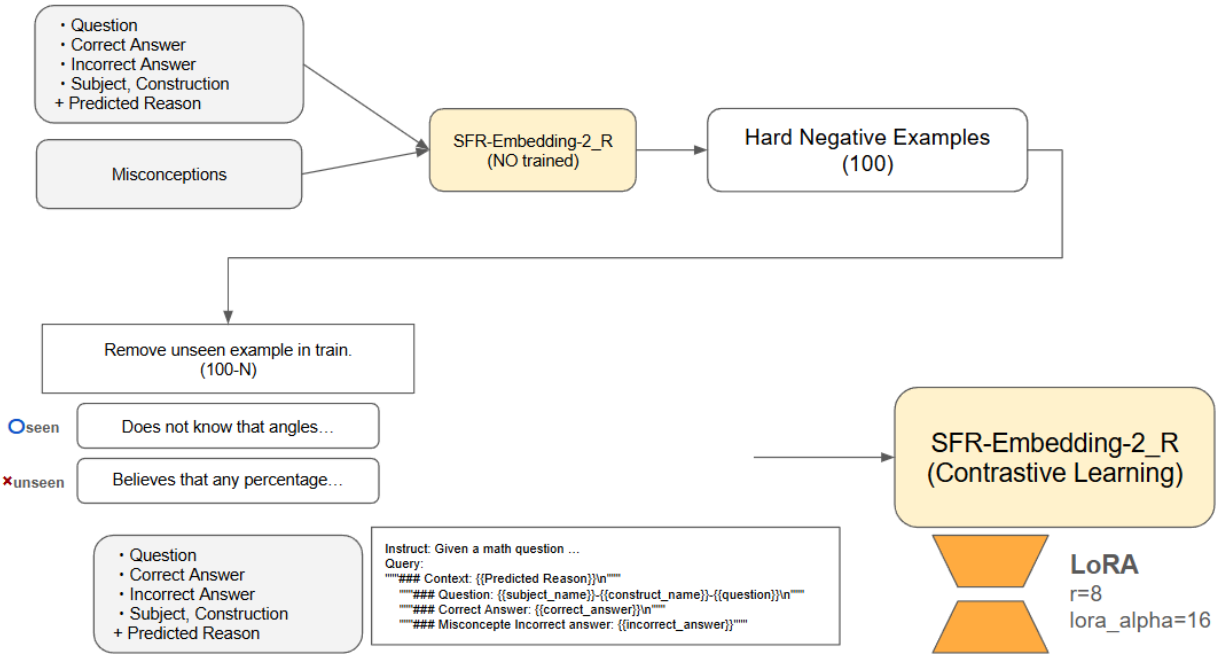
Merge Result

I used rerank model twice. First, I predicted 0~40 candidates, second I predicted 20~60. I merged these result, and picked top-20 from first and picked top5 from second.

This idea is mainly by ahmet.

Detailed

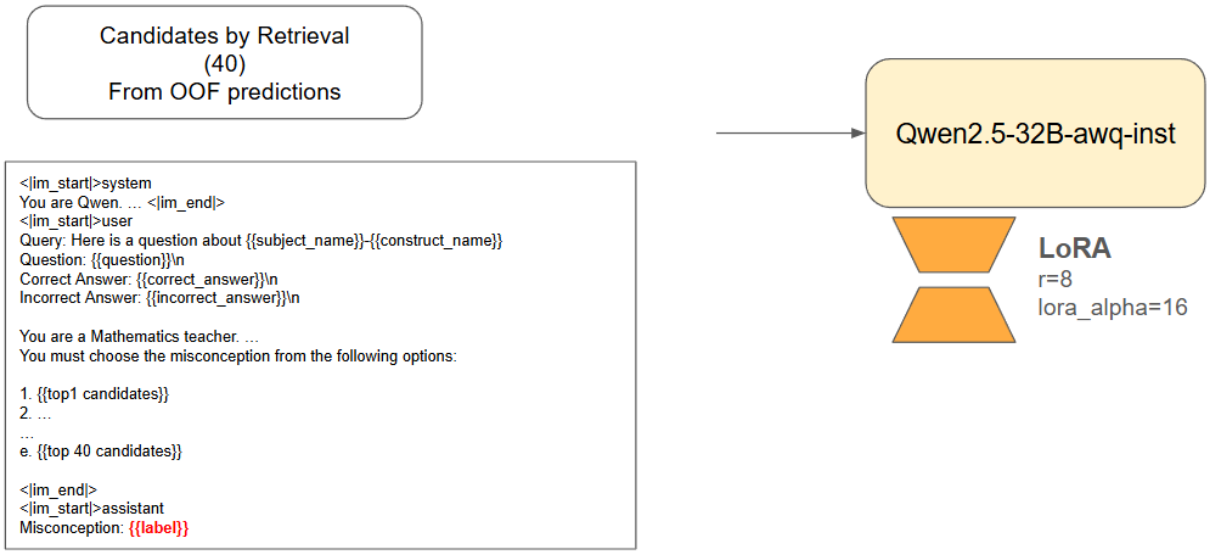
Retrieval



- We made the retriever model (Bi-Encoder model). High recall as well as high MAP is important in this phase.
- Improvement score points
  - Add the reason predicted by LLM (Qwen2.5-32B) as a context
  - Remove unseen example in train.
    - CV does not change, LB improved much
  - Fill LLM output into null misconception
  - LLM rephrasing about misconceptions

- Training
  - I mainly used it as a reference from this discussion
  - model : Salesforce/SFR-Embedding-2_R
    - with bnb-4bit
    - other models, such as qwen2.5, gte-qwen2 and bge-en-icl does not improve (or almost the same) in my experiments.
  - negative mining
    - num_negatives=4
    - seq_length
      - query=128
      - passage=64
  - LoRA
    - r=4
    - lora_alpha=8
  - epoch=15
  - lr=2e-05

Reranking



- I used LLM for reranking. I think that LLM is so effective in this competition due to LLM's having knowledge beforehand.
- prompt
  - I train LLM for predict misconception directly from multi-options. I need to design so that output token should be single token, so I make [1-9], [a-z] and [A-Z] precited token.
  - In training phase, only final token is made input as token classification.
  - In inference phase, final token is predicted by LLM.
  - For example

```
<|im_start|>system
You are Qwen, created by Alibaba Cloud. You are a helpful assistant.<|im_end|>
<|im_start|>user
Here is a question about Use the order of operations to carry out calculations involving
powers(BIDMAS).
Question: \[
3 \ 2+4-5
\]
Where  the brackets need to go to make the answer equal \( 13 \) ?
Correct Answer: \( 3 \(2+4)-5 \)
Incorrect Answer: \( 3 \ 2+(4-5) \)

You are a Mathematics teacher. Your task is to reason and identify the misconception behind the
Incorrect Answer with the Question.
You must choose the misconception from the following options:

1. Inserts brackets but not changed order of operation
2. Carries out operations from left to right regardless of priority order, unless brackets are used
3. Applies BIDMAS  strict order (does not realize addition and subtraction, and multiplication and
division, are of equal priority)
4. Carries out operations from left to right regardless of priority order
5. Answers order of operations questions with brackets as  the brackets are not there
6. Has not realised that the answer may be changed by the insertion of brackets
7. Carries out operations from right to left regardless of priority order
8. Does not include brackets when required
9. Performs addition ahead of any other operation
A. May have made a calculation error using the order of operations
B. Believes order of operations does not affect the answer to a calculation
C. Does not include brackets when attempting to multiply an expression
...
d. Believes that adding a negative will increase the value of the original number
e. When subtracting past zero, adds the absolute values and  writes the answer with a negative sign

<|im_end|>
<|im_start|>assistant
Misconception: 2
```

- Training
  - I used the retriever output of OOF prediction as options of prompt.
  - model: Qwen/Qwen2.5-32B-Instruct-awq
  - LoRA
    - r=8
    - lora_alpha=16
  - lr=1e-04
  - epoch=2

What didn’t work

so many… I wrote later